BRIO Research Paper

Phase 1: Understanding State Estimation

Detailed Explanation of the 15-Dimensional Drone State

Reference Paper:

A Robust Baro-Radar-Inertial Odometry M-Estimator for Multicopter Navigation in Cities and Forests

Authors:

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Paper Reference:

Section IV-A: System State

Equation (3), Page 3

Research Paper Reproduction Project

Phase 1 – Complete Study Notes

1 Introduction to State Estimation

1.1 What is State Estimation?

Imagine a drone flying in a forest.

To navigate safely, the drone needs to know:

- Where am I located?
- How fast am I moving?
- In which direction am I moving?
- How am I rotated?
- Are my sensor measurements accurate?

The process of estimating this information from sensor measurements is called **state estimation**.

Important Concept

State estimation is the process of calculating the drone's position, velocity, orientation, and sensor biases using available sensor measurements.

The drone does not directly know its exact state.

Instead, it receives measurements from sensors.

In the BRIO system, the primary sensors are:

1. Inertial Measurement Unit (IMU)
2. Radar
3. Barometer

These measurements are combined using factor-graph optimization.

The optimizer estimates the drone's state by finding values that best satisfy the available measurement constraints.

1.2 Example of State Estimation

Suppose the drone starts from:

$$P_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

After two seconds, suppose its actual position is:

$$P = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

Its velocity is:

$$V = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Its orientation is:

$$\phi = 0^\circ$$

$$\theta = 10^\circ$$

$$\psi = 0^\circ$$

where:

- ϕ is roll.
- θ is pitch.
- ψ is yaw.

This means the drone is:

- 4 meters along the X-axis.

- 2 meters above its starting position.
- Moving along X at 2 m/s.
- Moving upward at 1 m/s.
- Tilted by 10 degrees in pitch.

However, the drone does not directly know these exact values.

The estimator must calculate them from the sensor measurements.

2 What is a State?

A state is a collection of variables that describe the drone at a particular time.

For example:

$$x_0 = \text{Drone state at time index 0}$$

$$x_1 = \text{Drone state at time index 1}$$

$$x_2 = \text{Drone state at time index 2}$$

The subscript i represents a time index.

It does not necessarily represent time in seconds.

Each state contains the same types of variables, but their estimated values can change with time.

2.1 State Equation in the BRIO Paper

In Section IV-A, Equation (3), the paper defines the drone state as:

$$\boxed{x_i = \begin{bmatrix} R^i_{IB} \\ {}^{I}t^i_{IB} \\ {}^{I}v^i_{IB} \\ {}^{B}b^i_g \\ {}^{B}b^i_a \end{bmatrix}} \tag{1}$$

This equation contains five components:

1. Orientation
2. Position
3. Velocity
4. Gyroscope bias
5. Accelerometer bias

Important Concept

The BRIO system estimates a 15-dimensional drone state.
The state consists of five components, each contributing three independent degrees of freedom.

3 Understanding the Mathematical Notation

Before understanding each component, we must understand the mathematical symbols used in Equation (3).

3.1 Meaning of x_i

The symbol:

$$x_i$$

represents the drone state at time index i.

Here:

- x represents the state.
- i represents the time index.

For example:

$$x_1 = \text{State at the first indexed instant}$$

$$x_2 = \text{State at the second indexed instant}$$

The estimator creates a new state whenever a new radar frame arrives.

3.2 Meaning of Coordinate Frames

The paper uses three coordinate frames:

- I – Inertial frame
- B – Body frame
- R – Radar frame

For understanding Equation (3), we first focus on the inertial and body frames.

3.2.1 Inertial Frame – I

The inertial frame is a fixed reference coordinate system.

It contains three axes:

$$I = \{X_I, Y_I, Z_I\}$$

According to the paper:

- X_I and Y_I span a horizontal plane.
- Z_I points upward.

Imagine the drone begins its flight from:

$$P_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We can choose this as the origin of the inertial frame.

As the drone moves, the inertial frame remains fixed.

3.2.2 Body Frame – B

The body frame is attached to the drone.

According to the paper:

- X_B points forward.
- Y_B points left.
- Z_B points upward.

When the drone rotates, the body frame rotates with it.

Example

Imagine the drone initially faces north.
Its forward body axis points north.
Now the drone rotates 90 degrees to the right.
Its forward body axis points east.
The inertial frame remains fixed, but the body frame rotates with the drone.

3.3 Meaning of Superscripts and Subscripts

Consider:

$$^{I}t^{i}_{IB}$$

The notation contains several parts.

- t represents a translation vector.
- The superscript I indicates that the vector is expressed in the inertial frame.
- The subscript IB identifies the vector from the origin of frame I to the origin of frame B.
- The superscript i represents the time index.

Similarly:

$$^{B}b^{i}_{g}$$

means the gyroscope bias at time i, expressed in the body frame.

Remember

The coordinate-frame superscripts and subscripts describe how a vector is represented.
They are not multiplication operations.

4 First State Component: Orientation

The first term is:

$$\boxed{R^i_{IB}}$$

Here:

- R represents a rotation matrix.
- I represents the inertial frame.
- B represents the body frame.
- i represents the time index.

It describes the orientation of the drone's body frame relative to the inertial frame.

More precisely, it transforms vector coordinates from frame B to frame I.

4.1 What is Orientation?

Orientation describes how the drone is rotated in three-dimensional space.

A drone can rotate in three ways:

1. Roll
2. Pitch
3. Yaw

4.1.1 Roll

Roll describes rotation around the drone's forward axis.

For example:

$$\phi = 15^\circ$$

This indicates a sideways tilt.

4.1.2 Pitch

Pitch describes rotation around the drone's side-to-side axis.

For example:

$$\theta = 10^\circ$$

This indicates that the drone is tilted forward or backward.

4.1.3 Yaw

Yaw describes rotation around the drone's vertical axis.

For example:

$$\psi = 90^\circ$$

This represents a 90-degree heading rotation.

4.2 Example Orientation

Suppose:

$$\phi = 5^\circ$$

$$\theta = 10^\circ$$

$$\psi = 30^\circ$$

These values describe the drone's roll, pitch, and yaw.

The orientation has three independent degrees of freedom.

4.3 Why Does the Paper Use a Rotation Matrix?

Although orientation can be described using roll, pitch, and yaw, the paper represents orientation using a rotation matrix.

The rotation matrix is:

$$R_{IB} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

It contains nine matrix entries.

However, these entries are not independent.

A valid rotation matrix satisfies:

$$R^T R = I_3$$

and:

$$\det(R) = 1$$

where I_3 is the 3×3 identity matrix.

Therefore, a three-dimensional rotation has only three independent degrees of freedom.

Mathematically:

$$R_{IB} \in SO(3)$$

where $SO(3)$ is the set of valid three-dimensional rotation matrices.

Important Concept

A rotation matrix contains nine numerical entries, but only three independent degrees of freedom.
Therefore, orientation contributes three dimensions to the drone state.

4.4 Why Are Rotation Matrices Useful?

Rotation matrices are useful because they:

1. Transform vectors between coordinate frames.
2. Allow rotations to be combined using matrix multiplication.
3. Avoid the singularities associated with certain Euler-angle representations.

In BRIO, they are particularly important because radar and IMU measurements are represented in different coordinate frames.

4.5 Example: Drone Rotates 90 Degrees

Suppose the drone moves forward at:

$$^{B}v = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \text{ m/s}$$

This means the drone moves forward at 2 m/s in its body frame.

Now suppose the drone rotates 90 degrees counterclockwise around the vertical axis.

The rotation matrix becomes:

$$R_{IB} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The inertial-frame velocity is:

$$^{I}v = R_{IB}{}^{B}v$$

Substituting:

$$^{I}v = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

After multiplication:

$$\boxed{^{I}v = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}}$$

Example

Before rotation, the drone moves along the inertial X-axis.
After rotation, the drone's forward movement is along the inertial Y-axis.
Its speed remains 2 m/s.
Only its direction changes.

4.6 Rotation Composition

The paper uses multiple coordinate frames.

Let:

$$R_{IB}$$

represent the rotation from body frame to inertial frame.

Let:

$$R_{BR}$$

represent the rotation from radar frame to body frame.

Then:

$$\boxed{R_{IR} = R_{IB}R_{BR}}$$

This relates the radar frame directly to the inertial frame.

The relationship follows Equation (2) of the paper.

Remember

The rotation matrix allows measurements from different coordinate frames to be combined.

5 Second State Component: Position

The second term is:

$$\boxed{{}^{I}t^{i}_{IB}}$$

It represents the drone's position in the inertial frame at time i.

5.1 Understanding the Symbols

- t – Translation vector.
- Superscript I – Expressed in the inertial frame.
- Subscript IB – Vector from the inertial origin to the body-frame origin.
- Superscript i – Time index.

5.2 Mathematical Representation

The drone position is:

$$\boxed{{}^{I}t^{i}_{IB} = \begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix}}$$

where:

- x_i is the X-position.
- y_i is the Y-position.
- z_i is the height.

5.3 Position Example

Suppose:

$$ {}^{I}t^{i}_{IB} = \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} $$

This means:

- X-position = 5 meters.
- Y-position = 3 meters.
- Z-position = 2 meters.

The drone is located at:

$$ (5, 3, 2) $$

relative to the chosen inertial origin.

Remember

Position answers the question:
Where is the drone located?
Position contributes three dimensions.

6 Third State Component: Velocity

The third term is:

$$ \boxed{{}^{I}v^{i}_{IB}} $$

It represents the drone's linear velocity in the inertial frame.

6.1 Understanding the Symbols

- v – Linear velocity.
- Superscript I – Expressed in the inertial frame.

- Subscript IB – Motion of the body-frame origin relative to the inertial frame.
- Superscript i – Time index.

6.2 Velocity Representation

$$\boxed{{}^{I}v_{IB}^{i} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}}$$

where:

- v_x is velocity along X.
- v_y is velocity along Y.
- v_z is velocity along Z.

6.3 Velocity Example

Suppose:

$${}^{I}v_{IB}^{i} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \text{ m/s}$$

This means:

- The drone moves along X at 2 m/s.
- The drone has zero Y velocity.
- The drone moves upward at 1 m/s.

6.4 Difference Between Position and Velocity

Position tells us where the drone is.

Velocity tells us how its position changes over time.

Consider a simple example.

At time $t = 0$:

$$P_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Suppose the drone moves at constant velocity:

$$V = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \text{ m/s}$$

After one second:

$$P_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

After two seconds:

$$P_2 = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$$

After three seconds:

$$P_3 = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$$

The velocity remains constant, but the position changes.

For constant velocity:

$$\boxed{P_{\text{new}} = P_{\text{old}} + V\Delta t}$$

where Δt is the elapsed time.

This is a simplified motion model.

The actual state estimator also accounts for acceleration, orientation, sensor biases, and uncertainty.

6.5 How Does BRIO Estimate Velocity?

The IMU measures acceleration, which is used to predict changes in velocity.

The radar measures radial Doppler velocity along the directions of detected objects.

The radar measurements constrain the drone's velocity estimate.

A single Doppler measurement does not directly provide the complete three-dimensional velocity.

In principle, three linearly independent radar detection directions are sufficient to constrain three-dimensional linear velocity.

The estimator combines radar observations with IMU-based motion predictions.

Remember

Velocity answers:

How fast and in which direction is the drone moving?

Velocity contributes three dimensions.

7 Fourth State Component: Gyroscope Bias

The fourth term is:

$$\boxed{{}^{B}b_{g}^{i}}$$

It represents the gyroscope bias at time i, expressed in the body frame.

7.1 What is Bias?

Bias is a systematic measurement offset.

Real sensors are not perfect.

They may produce slightly incorrect measurements even when the physical quantity is known.

7.2 Gyroscope Bias Example

Imagine the drone is stationary.

Its actual angular velocity is:

$$\omega_{\text{true}} = 0^\circ/s$$

However, the gyroscope reports:

$$\omega_{\text{measured}} = 0.5^\circ/s$$

Ignoring random noise, the gyroscope bias is:

$$b_g = 0.5^\circ/s$$

A simplified gyroscope measurement model is:

$$\boxed{\omega_{\text{measured}} = \omega_{\text{true}} + b_g + n_g}$$

where:

- ω_{measured} is the measured angular velocity.
- ω_{true} is the actual angular velocity.
- b_g is the gyroscope bias.
- n_g is measurement noise.

7.3 Why is Gyroscope Bias Dangerous?

Suppose a stationary gyroscope has a constant bias:

$$b_g = 0.5^\circ/s$$

If the estimator does not correct it, the orientation error can accumulate.

After 100 seconds:

$$\Delta\theta = b_g t$$

$$\Delta\theta = 0.5 \times 100$$

$$\boxed{\Delta\theta = 50^\circ}$$

The drone has not rotated, but the estimated orientation could drift by 50 degrees.

This example assumes constant bias and simple single-axis integration.

7.4 Three Gyroscope Bias Variables

The gyroscope measures angular velocity around three axes.

Therefore:

$$\boxed{{}^{B}b_g^i = \begin{bmatrix} b_{gx} \\ b_{gy} \\ b_{gz} \end{bmatrix}}$$

For example:

$${}^{B}b_g^i = \begin{bmatrix} 0.1 \\ -0.2 \\ 0.5 \end{bmatrix}^\circ /s$$

Each gyroscope axis can have a different measurement offset.

Therefore, gyroscope bias contributes three dimensions.

7.5 Gyroscope Bias in the BRIO Paper

In Section IV-H, the authors explain that yaw-related gyroscope bias is difficult to observe accurately using their radar-inertial configuration.

Therefore, they perform turn-on-bias calibration.

The initial gyroscope readings are averaged to estimate the bias.

The estimated initial bias is then initialized with high certainty.

This helps reduce orientation drift.

Remember

Gyroscope bias is estimated because incorrect angular velocity measurements can cause orientation drift.

Gyroscope bias contributes three dimensions.

8 Fifth State Component: Accelerometer Bias

The fifth term is:

$$\boxed{{}^{B}b_a^i}$$

It represents the accelerometer bias at time i, expressed in the body frame.

8.1 What is Accelerometer Bias?

An accelerometer measures specific force.

Its readings are used together with orientation and gravity to estimate the drone's translational acceleration.

Real accelerometers also contain systematic measurement errors.

These errors are called accelerometer biases.

8.2 Accelerometer Bias Example

Suppose the ideal accelerometer measurement along one axis is:

$$f_{\text{true},x} = 0$$

But the accelerometer reports:

$$f_{\text{measured},x} = 0.1 \text{ m/s}^2$$

Ignoring random noise:

$$b_{ax} = 0.1 \text{ m/s}^2$$

A simplified accelerometer model is:

$$\boxed{f_{\text{measured}} = f_{\text{true}} + b_a + n_a}$$

where:

- f_{measured} is measured specific force.
- f_{true} is ideal specific force.
- b_a is accelerometer bias.
- n_a is measurement noise.

Important Concept

An accelerometer at rest does not necessarily report zero on all axes.
It measures specific force, which includes the effect of gravity in the accelerometer measurement model.

8.3 How Does Accelerometer Bias Affect Velocity?

Consider a simplified one-dimensional example.

Suppose an uncorrected accelerometer bias is:

$$b_a = 0.1 \text{ m/s}^2$$

The drone is actually stationary.

However, the estimator mistakenly interprets the bias as acceleration.

After ten seconds:

$$\Delta v = at$$

$$\Delta v = 0.1 \times 10$$

$$\boxed{\Delta v = 1 \text{ m/s}}$$

The estimated velocity can develop a 1 m/s error.

8.4 How Does Accelerometer Bias Affect Position?

For constant acceleration:

$$\Delta p = \frac{1}{2} a t^2$$

Substituting:

$$\Delta p = \frac{1}{2}(0.1)(10)^2$$

$$\boxed{\Delta p = 5 \text{ m}}$$

Therefore, an accelerometer bias can produce significant position drift.

This is a simplified example that ignores rotation, gravity-estimation errors, and measurement corrections.

8.5 Three Accelerometer Bias Variables

Accelerometer bias is represented as:

$$\boxed{{}^{B}b_a^i = \begin{bmatrix} b_{ax} \\ b_{ay} \\ b_{az} \end{bmatrix}}$$

For example:

$${}^{B}b_a^i = \begin{bmatrix} 0.1 \\ 0.02 \\ -0.05 \end{bmatrix} \text{ m/s}^2$$

Accelerometer bias contributes three dimensions.

Remember

Accelerometer bias is estimated because acceleration errors can accumulate into velocity and position errors.
Accelerometer bias contributes three dimensions.

9 Why Does the State Have 15 Dimensions?

We have now studied all five components.

Component	Meaning	Dimensions
Orientation	Drone rotation	3
Position	Drone location	3
Velocity	Drone linear motion	3
Gyroscope bias	Angular velocity offset	3
Accelerometer bias	Acceleration measurement offset	3
Total	**Drone state**	**15**

Therefore:

$$\boxed{3+3+3+3+3=15}$$

Although orientation is represented using a 3×3 rotation matrix, it has only three independent degrees of freedom.

This is why the complete state is called 15-dimensional.

10 Example of a Complete Drone State

Suppose the optimizer estimates the state at a particular time.

The values might be:

$$P = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} \text{ m}$$

$$V = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \text{ m/s}$$

For illustration, describe the orientation using Euler angles:

$$\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} = \begin{bmatrix} 0^\circ \\ 10^\circ \\ 0^\circ \end{bmatrix}$$

The gyroscope bias is:

$$b_g = \begin{bmatrix} 0.01 \\ 0.02 \\ 0.03 \end{bmatrix}^{\circ} /s$$

The accelerometer bias is:

$$b_a = \begin{bmatrix} 0.1 \\ 0.02 \\ 0.05 \end{bmatrix} \text{ m/s}^2$$

These five components together describe the estimated state.

The numerical values are illustrative, not experimental results from the paper.

11 How Sensors Contribute to State Estimation

The BRIO system combines measurements from three physical sensors:

1. IMU
2. Radar
3. Barometer

Each sensor contributes different information.

11.1 IMU

The IMU contains:

- Gyroscope
- Accelerometer

The gyroscope measures angular velocity.

The accelerometer measures specific force.

IMU measurements help predict changes in:

- Orientation

- Velocity
- Position

However, IMU measurements contain noise and bias.

Therefore, IMU-only navigation can accumulate drift.

11.2 Radar

The radar provides detections containing quantities such as:

- Direction of arrival
- Doppler velocity
- Range
- Signal-to-noise ratio
- Detection noise information

In the BRIO estimator, the bearing Doppler factors primarily use detection direction and Doppler velocity.

These constrain the drone's motion.

The radar does not simply output the complete 15-dimensional state.

Instead, individual radar detections contribute measurement constraints.

11.3 Barometer

The barometer measures atmospheric pressure.

Pressure measurements are converted into height-related measurements using an atmospheric model.

The barometer helps reduce vertical position drift.

The paper uses a robust barometric factor to reduce the influence of disturbances.

11.4 Sensor Contribution Summary

Sensor	Measurement	Contribution
Gyroscope	Angular velocity	Orientation changes
Accelerometer	Specific force	Velocity and position prediction
Radar	Doppler velocity and direction	Velocity constraints
Barometer	Atmospheric pressure	Vertical position constraints

Important Concept

Sensors do not independently calculate the complete drone state.
They provide measurements.
The optimizer combines the measurement constraints to estimate the state.

12 State Estimation at Different Time Instances

The paper creates one state for each radar frame.

The experimental radar operates at approximately 8 Hz.

Therefore, the nominal interval is:

$$\Delta t = \frac{1}{8}$$

$$\boxed{\Delta t = 0.125 \text{ s}}$$

For illustration, consider:

$$t_0 = 0$$

$$t_1 = 0.125 \text{ s}$$

$$t_2 = 0.250 \text{ s}$$

The estimator creates:

$$x_0, \quad x_1, \quad x_2$$

Each state contains 15 dimensions.

For three states:

$$15 \times 3 = 45$$

Thus, the three drone states have 45 independent state dimensions.

12.1 Example of Three States

At the first state:

$$P_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$V_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

At the second state:

$$P_1 = \begin{bmatrix} 0.01 \\ 0 \\ 0 \end{bmatrix}$$

$$V_1 = \begin{bmatrix} 0.2 \\ 0 \\ 0 \end{bmatrix}$$

At the third state:

$$P_2 = \begin{bmatrix} 0.05 \\ 0 \\ 0 \end{bmatrix}$$

$$V_2 = \begin{bmatrix} 0.4 \\ 0 \\ 0 \end{bmatrix}$$

These are illustrative values showing how states change over time.

They are not intended as exact numerical integration results.

Each state also contains:

- Orientation
- Gyroscope bias
- Accelerometer bias

13 Set of Estimated States

The paper defines the set of estimated drone states as:

$$\boxed{X_k = \{x_i\}_{i \in K_k}} \tag{2}$$

This is part of Equation (4) in the paper.

Here:

- X_k is the set of estimated states up to time index k.
- x_i is the state at index i.
- K_k is the set of radar-frame indices up to index k.

For example:

$$X_2 = \{x_0, x_1, x_2\}$$

The optimizer estimates multiple connected states.

In the practical implementation, a sliding optimization window is used to keep computation manageable.

13.1 Additional Tracked Radar Positions

The paper also estimates positions of certain tracked stationary radar detections.

These tracked positions are written as:

$$t^j$$

The set of tracked positions is:

$$T_k = \{t^j\}_{j \in L_k}$$

These are additional optimization variables.

They are not included in the 15-dimensional drone state.

14 What Does the Optimizer Calculate?

The optimizer calculates the drone states that best satisfy the available sensor measurements.

The measurements contain uncertainty.

For example:

- IMU measurements contain bias and noise.
- Radar detections may include moving objects or ghost targets.
- Barometer readings may be disturbed by aerodynamic effects.

Therefore, the estimator must combine the measurements carefully.

14.1 Simplified Velocity Example

Suppose an IMU-based motion prediction suggests:

$$v_x = 2.2 \text{ m/s}$$

Radar observations constrain the velocity toward:

$$v_x = 2.0 \text{ m/s}$$

The optimizer considers measurement uncertainties and the other state constraints.

It might estimate:

$$\boxed{v_x = 2.05 \text{ m/s}}$$

This is only an illustrative example.

The actual result depends on the measurement residuals, uncertainties, robust loss functions, and other estimated variables.

14.2 What is the Optimizer Output?

For each state x_i, the optimizer estimates:

$$\boxed{x_i = \begin{bmatrix} \text{Orientation} \\ \text{Position} \\ \text{Velocity} \\ \text{Gyroscope bias} \\ \text{Accelerometer bias} \end{bmatrix}}$$

The optimizer does not simply select the output of one sensor.

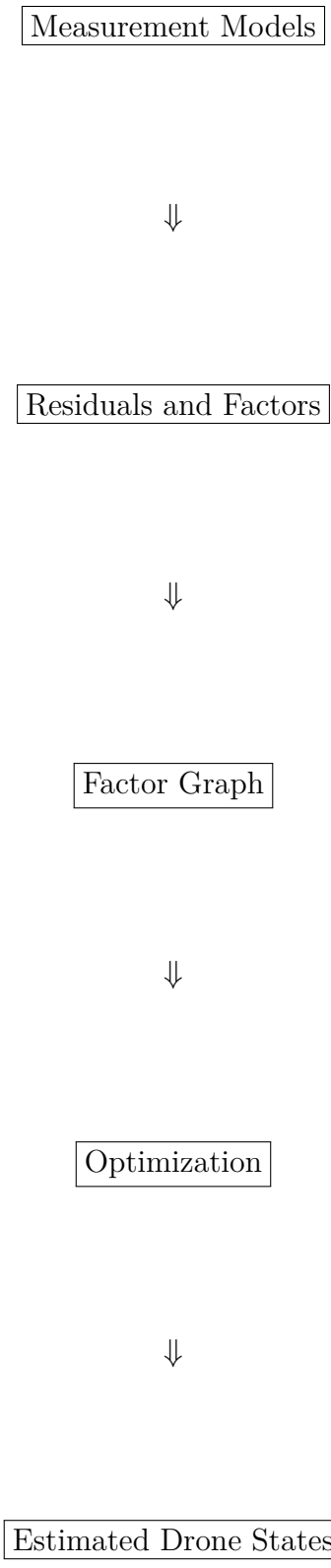
Instead, it searches for a mutually consistent set of states.

15 Overall Phase 1 Flow

The conceptual process is:

Sensor Measurements

$\Downarrow$



The estimated states contain:

$$\boxed{R^i_{IB},\quad {}^I t^i_{IB},\quad {}^I v^i_{IB},\quad {}^B b^i_g,\quad {}^B b^i_a}$$

Important Concept

The factor graph represents the measurement constraints and their relationships to the state variables.
The optimizer uses this graph to calculate the most consistent state estimates.
The factor graph is not a separate processing stage that receives values only after optimization.

16 Phase 1: Important Questions

16.1 Question 1

Why does the paper estimate 15 dimensions?

Because the state includes:

- 3 orientation degrees of freedom
- 3 position variables
- 3 velocity variables
- 3 gyroscope bias variables
- 3 accelerometer bias variables

Thus:

$$3 + 3 + 3 + 3 + 3 = 15$$

16.2 Question 2

What is the difference between position and velocity?

Position tells us where the drone is located.

Velocity tells us how quickly and in what direction the position changes.

16.3 Question 3

Why do we estimate gyroscope bias?

Because gyroscope bias can accumulate into orientation drift.

Correcting or accounting for the bias improves orientation estimation.

16.4 Question 4

Why do we estimate accelerometer bias?

Because accelerometer bias can produce velocity and position errors when acceleration measurements are integrated over time.

16.5 Question 5

Why is orientation important for radar and IMU fusion?

The radar and IMU operate in coordinate frames attached to the drone.

The drone's orientation is needed to relate these measurements to the inertial frame.

16.6 Question 6

Why use a 3×3 rotation matrix instead of only three angles?

Rotation matrices make coordinate transformations and rotation composition convenient and mathematically consistent.

Although they contain nine entries, they have only three independent degrees of freedom.

16.7 Question 7

Does each sensor independently calculate the entire state?

No.

Each sensor provides measurements that constrain different components of the state.

The optimizer combines all available constraints.

17 Final Phase 1 Summary

Concept	Meaning
State estimation	Estimating drone motion from sensor measurements
State	Information describing the drone at one time
Position	Where the drone is located
Velocity	How the drone moves
Orientation	How the drone is rotated
Gyroscope bias	Angular velocity measurement offset
Accelerometer bias	Accelerometer measurement offset
Rotation matrix	Transforms vectors between coordinate frames
15-dimensional state	Five components with three dimensions each
Factor graph	Represents relationships between states and measurements
Optimizer	Finds state values that best satisfy measurement constraints

Remember

The most important concept of Phase 1:
The sensors do not directly provide the complete drone state.
They provide measurements.
The estimator combines these measurements through mathematical constraints and optimization.
The final output is the estimated 15-dimensional drone state at each radar frame.

18 Next Phase

The next phase is:

Phase 2: Understanding the Sensors and Coordinate Frames

Topics include:

1. Inertial frame
2. Body frame

3. Radar frame
4. Rotation matrices
5. Rigid transformations
6. IMU measurements
7. Radar measurements
8. Barometer measurements
9. Measurement timestamps

These concepts are necessary to understand how the BRIO system combines measurements from multiple sensors.

References

[1] R. Girod, M. Hauswirth, P. Pfreundschuh, M. Biasio, and R. Siegwart,

A Robust Baro-Radar-Inertial Odometry M-Estimator for Multicopter Navigation in Cities and Forests ,

2024.

arXiv:2408.05764.

`https://arxiv.org/abs/2408.05764`